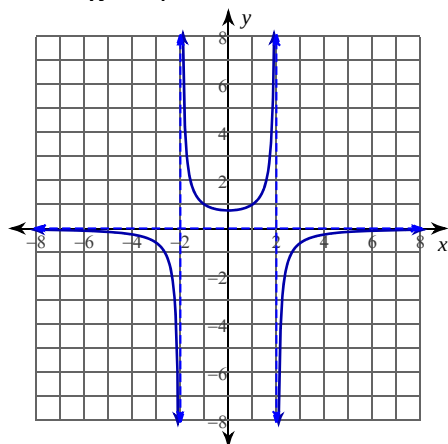
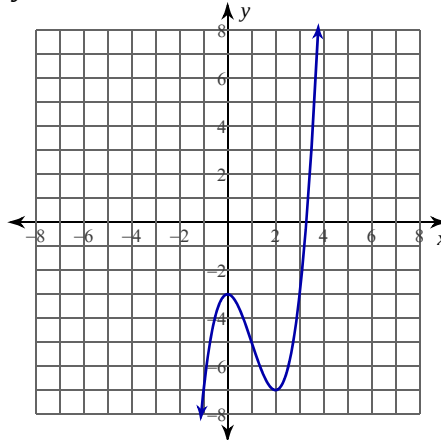


Approximate the relative and absolute extrema of each function. Then approximate the intervals where each function is increasing and decreasing.

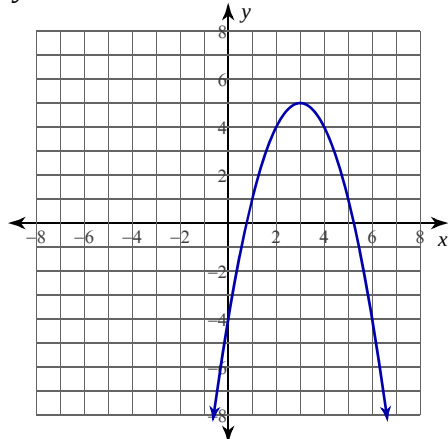
9) $y = -\frac{3}{x^2 - 4}$



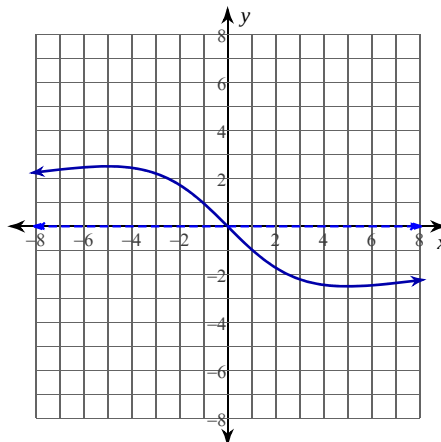
10) $y = x^3 - 3x^2 - 3$



11) $y = -x^2 + 6x - 4$



12) $y = -\frac{25x}{x^2 + 25}$



Critical thinking questions:

13) Write a function that has the following relative maximums: $(1, 1)$, $(2, 2)$, $(3, 3)$.

14) Is it possible for a continuous function to have only the following extrema?

Relative max: $(1, 1)$, $(3, 3)$

Relative min: $(2, 2)$

Explain why or why not.